

OLS Regression Results

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Dep. Variable:          y      R-squared:          0.891
Model:                  OLS    Adj. R-squared:       0.890
Method:                 Least Squares    F-statistic:       607.6
Date:                  Tue, 23 Jan 2024    Prob (F-statistic): 2.04e-37
Time:                  22:20:19    Log-Likelihood:    -102.30
No. Observations:      76      AIC:              208.6
Df Residuals:          74      BIC:              213.3
Df Model:               1
Covariance Type:       nonrobust
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              coef      std err          t      P>|t|      [0.025      0.975]
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const         3.2028         0.221     14.499      0.000         2.763         3.643
x              9.1104         0.370     24.650      0.000         8.374         9.847
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Omnibus:          5.816    Durbin-Watson:
Prob(Omnibus):    0.055    Jarque-Bera (JB):
Skew:             -0.112   Prob(JB):
Kurtosis:         2.126    Cond. No.
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                                1.896
                                2.579
                                0.275
                                4.42
=====
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Handwritten annotations: A green circle around "Durbin-Watson:" and "4.42". A green circle around "2.579". A green arrow pointing to "1.896". A green checkmark and a green question mark next to the bottom two values.

We have already encountered some of the generated numbers like R², F-statistic, etc. But the adjacent block of our output contains many others ... what are they, why are they important, and how to interpret them?

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Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
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This is the output generated by the OLS function on a sample dataset

AIC (Akaike Information Criteria) and BIC (Bayes Information ...)

- Both these are estimators of **prediction error**. They help in model selection.
- These numbers are used to compare across models. **A lower number indicates a better model**
- A difference in AIC or BIC value of 2, between models being compared, is considered significant. The model with a lower AIC or BIC value is designated as the better model, and becomes a candidate for selection.

- **Omnibus statistic:** This is a numeric value calculated from the skewness and kurtosis of the residuals (the difference between the predicted and actual values). A low value suggests the residuals are closer to a normal distribution, while a high value indicates deviation from normality.
- **Omnibus p-value:** This value represents the probability of observing the calculated Omnibus statistic, assuming the null hypothesis of normally distributed residuals is true. A low p-value (typically < 0.05) suggests there is significant evidence to reject the null hypothesis, implying the residuals are not normally distributed.

Skewness measures the asymmetry of the probability distribution of a dataset. For a normal distribution, the skewness should be **close to 0**.

- **Skewness = 0:** The data is symmetric, which is a characteristic of a normal distribution.
- **Skewness > 0:** The data is positively skewed (right-tailed), meaning the tail on the right side is longer or fatter.
- **Skewness < 0:** The data is negatively skewed (left-tailed), meaning the tail on the left side is longer or fatter.

Interpretation:

- If the skewness is significantly different from 0, the data is unlikely to follow a normal distribution.

Kurtosis measures the "tailedness" of the probability distribution, indicating whether the data has heavy or light tails compared to a normal distribution. For a normal distribution, the kurtosis is **close to 3** (or 0 if using excess kurtosis).

- **Kurtosis = 3 (or Excess Kurtosis = 0):** The data has tails similar to a normal distribution.
- **Kurtosis > 3 (or Excess Kurtosis > 0):** The data has heavier tails (more outliers) than a normal distribution (leptokurtic).
- **Kurtosis < 3 (or Excess Kurtosis < 0):** The data has lighter tails (fewer outliers) than a normal distribution (platykurtic).

Interpretation:

- If the kurtosis is significantly different from 3 (or excess kurtosis significantly different from 0), the data is unlikely to follow a normal distribution.

Jarque-Bera Test

- In the context of Ordinary Least Squares (OLS) regression, the Jarque-Bera test is used to **check the normality of the residuals**.
 - **Residuals**, the difference between the predicted and actual values in your model, play a crucial role in OLS analysis.
 - Their normality is one of the key assumptions for the validity of statistical inferences drawn from the model.
- A **low statistic** : Low value of the Jarque-Bera statistic (< 2) along with high p-value (ie. > 0.05) indicate that the residuals follow Normal Distribution.
- A **high statistic** : High value of the Jarque-Bera statistic (> 6) often accompanied with low p-values (ie. < 0.05) indicate that the residuals DO NOT follow Normal Distribution.

Durbin-Watson Test

- In the context of Ordinary Least Squares (OLS) regression, the Durbin-Watson (DW) test is a diagnostic tool used to check for **autocorrelation** in the residuals (errors) of the model. Autocorrelation occurs when there's a dependence between subsequent errors, meaning the error term at one point in time influences the error term at another point.
- Its value always falls between 0 and 4, with specific interpretations:
 - **2.0:** Indicates no autocorrelation (ideal scenario).
 - **0 to less than 2.0:** Suggests positive autocorrelation (errors tend to cluster together, either positive or negative).
 - **More than 2.0 to 4:** Suggests negative autocorrelation (errors tend to alternate between positive and negative).
- This test is used as a first check, and not a definitive test.

Condition Number

The Condition Number in Ordinary Least Squares (OLS) refers to a **measure of how sensitive the estimated coefficients are to small changes in the data**. It's not directly related to any specific variable or error term, but rather evaluates the overall stability and robustness of the model's solution. Its calculation is based on eigenvalues

- A **low Condition Number** indicates that the coefficients react minimally to small changes in the data (stable, robust model).
- A **high Condition Number** signifies that even slight data variations can significantly alter the coefficients (sensitive, potentially unstable model).

Why is it important?

- A high Condition Number suggests the model might be fitting noise or capturing spurious relationships due to its sensitivity to slight data changes. This makes the estimated coefficients less reliable and conclusions less trustworthy.
- In extreme cases, a very high Condition Number can lead to numerical issues during calculations, rendering the model estimation altogether unstable.

Interpretation:

There's no single threshold for a "good" or "bad" Condition Number. However, in general:

- **Values below 10** are considered acceptable, indicating a relatively stable model.
- **Values above 30** raise concerns about sensitivity and potential instability.
- **Values above 100** are a strong indicator of an unreliable model requiring further investigation or improvement.

Output	Interpretation	Specific Limits/Values (if applicable)
R^2	Proportion of variance in the dependent variable explained by the model.	Range: [0, 1], higher values are desirable.
Adjusted R^2	R^2 adjusted for the number of predictors; a measure of model fit.	Like R^2, but adjusted for model complexity.
F-statistic	Tests the overall significance of the regression model.	Critical values based on significance level (e.g., 0.05).
AIC (Akaike's IC)	A measure of model goodness-of-fit, balancing complexity and fit.	Lower values are better; used for model comparison.
BIC (Bayesian IC)	Similar to AIC but penalizes model complexity more heavily.	Lower values are better; stricter penalty for complexity.
Log Likelihood	A measure of how well the model explains the observed data.	Higher values indicate better model fit.
Omnibus	Refers to a specific statistic and its associated p-value that test the normality of the residuals. It is a combination of multiple tests like Jarques-Bera test, Shapiro-Wilkes test and Kolmongoriv-Smirnov test.	For the residuals to have Normal Distribution, the Omnibus statistic should have low value and p-value should be > 0.05
Durbin-Watson test	Tests for autocorrelation in the residuals; values around 2 suggest no autocorrelation.	Range: [0, 4], close to 2 indicates no significant autocorrelation.
Jarque-Bera test	Tests for normality of residuals based on skewness and kurtosis. Test statistic value closer to zero implies residuals are normally distributed NULL Hypothesis: The residuals are Normally Distributed	Critical values based on significance level (e.g., 0.05). If p-value < 0.05, then NULL Hypothesis is rejected implying the residuals are NOT normally distributed. If residuals are normally distributed, p-value > 0.05.
Condition Number	Measures sensitivity to changes in input variables; high values indicate multicollinearity.	No strict limits; values above 30 may indicate multicollinearity.
Skew	A measure of the asymmetry of the residuals distribution.	Range: $(-\infty, \infty)$; 0 for a perfectly symmetric distribution.
Kurtosis	A measure of the "tailedness" of the residuals distribution.	Range: $(-\infty, \infty)$; 3 for a normal distribution (excess kurtosis).